IoT and Fog-Computing-Based Predictive Maintenance Model for Effective Asset Management in Industry 4.0 Using Machine Learning

Yyi Kai Teoh, Sukhpal Singh Gill, and Ajith Kumar Parlikad, *Member, IEEE*

***Abstract*—The assets in Industry 4.0 are categorized into physical, virtual, and human. The innovation and popularization of ubiquitous computing enhance the usage of smart devices: RFID tags, QR codes, LoRa tags, etc., for asset identification and tracking. The generated data from the Industrial Internet of Things (IIoT) ease information visibility and process automation in Industry 4.0. Virtual assets include the data produced from IIoT. One of the applications of the industrial big data is to predict the failure of the manufacturing equipment. Predictive maintenance enables the business owner to decide, such as repairing or replacing the component before an actual failure that affects the whole production line. Therefore, Industry 4.0 requires an effective asset management to optimize the task distributions and predictive maintenance model. This article presents the genetic algorithm (GA)-based resource management integrating with machine learning for predictive maintenance in fog computing. The time, cost, and energy performance of GA along with MinMin, MaxMin, FCFS, and RoundRobin are simulated in the FogWorkflowsim. The predictive maintenance model is built in two-class logistic regression using real-time data sets. The results demonstrate that the proposed technique outperforms MinMin, MaxMin, FCFS, RoundRobin in execution time, cost, and energy usage. The execution time is 0.48% faster, 5.43% lower cost and energy usage is 28.10% lower in comparison with second-best results. The training and testing accuracy of the prediction model is 95.1% and 94.5%, respectively.**

***Index Terms*—Fog computing, industry 4.0, Internet of Things (IoT), predictive maintenance, resource management.**

I. Introduction

THE EXPONENTIAL growth of new-generation computing, such as cloud edge computing, Internet of Things (IoT), big data, cyber–physical system (CPS), etc., contributes substantially in the manufacturing industry to accomplish a more proficient, competing, and smart manufacturing. Smart manufacturing represents as a future condition of manufacturing, where the ongoing transmission and analysis of data from the shop floor produce manufacturing knowledge, which has a positive effect over all parts of activities. The Industrial IoT (IIoT) is an extension of the IoT to use in industrial sectors. IIoT mainly gathers the massively interconnected sensors industrial data at the shop floor to produce information, knowledge, and control manufacturing system [1]. The utilization of IIoT incorporates infrastructure, maintenance, process control, and supply chain.

GE intelligent platform reports that a healthcare product manufacturer creates 5000 samples each 33 ms, likeness 4 trillion of samples each year [2]. A manufacturing factory with 100 machine tools and ten cameras generates 72 TB of data per year [3]. Conventional in-house servers with constraining resources, i.e., storage, memory, and processing power are not fit for processing the new challenge due to scalability and computational complexity and shall deploy in the cloud data center. However, simulation research in [4] concludes that the cloud data center potentially experiences higher latency and network usage due to the vast geographical distance between IIoT devices and cloud data center. Fog computing as an expansion of cloud computing to the edge of system networking consists of cloud and edge resources that reduce the latency and network congestion. Therefore, latency-sensitive applications can be executed in fog computing.

A distributed system refers to multiple systems that are interlinked while appearing as a single system to the user to enhance resource sharing [5]. Resource management in a distributed system is a fundamental process that involves resource scheduling and allocating resources to applications [6]. The genetic algorithm (GA) has been generally applied to improve and optimize the resources allocation, and GA is one of the most dependable and promising metaheuristics [7]. GA is a part of the evolutionary algorithm, where it is inspired by the evolutionary theory and nature process selection.

The assets in the manufacturing sector include physical asset, human asset, and virtual asset. The reduction of tag cost and infrastructure cost greatly enhance the usage of RFID, 1-D barcodes, QR codes, BLE tag, and LoRa tag in asset management. Open platform communications united architecture (OPC UA) is the primary data exchange standard for industrial communication recommended by the Reference Architecture

Manuscript received 7 October 2020; revised 9 December 2020 and 31 December 2020; accepted 6 January 2021. Date of publication 11 January 2021; date of current version 24 January 2023. *(Corresponding author: Yyi Kai Teoh.)*

Yyi Kai Teoh and Sukhpal Singh Gill are with the School of Electronic Engineering and Computer Science, Queen Mary University of London, London E1 4NS, U.K. (e-mail: y.teoh@se19.qmul.ac.uk; s.s.gill@qmul.ac.uk).

Ajith Kumar Parlikad is with the Institute for Manufacturing, University of Cambridge, Cambridge CB3 0FS, U.K. (e-mail: aknp2@cam.ac.uk).

Digital Object Identifier 10.1109/JIOT.2021.3050441

Model for Industry 4.0 (RAMI 4.0). OPC UA allows the primary physical assets (manufacturing equipment) to communicate with each other or to exchange information with the gateway.

A. Motivation and Our Contributions

Industry 4.0 enhances the productivity of manufacturing technologies through collection and analysis of real-time data. The ease of communication in IIoT enables real-time tracking and identification of assets within businesses originating from IIoT sources and information services. Industrial big data generated from IIoT sensors promote information visibility. One of the critical applications in manufacturing is to predict the condition of manufacturing equipment. Therefore, there is a requirement of an efficient and effective resource management technique to handle the generated data. To this end, GA seems as a promising approach as GA is favorable in band selection, smaller size classification, training, and testing accuracies compared to the existing works.

The *main contributions* of this research article are as follows.

1) *Proposed Resource Scheduling Technique:* GA for asset management in Industry 4.0.
2) Simulated GA with various scheduling techniques, namely, MinMin, MaxMin, FCFS, and RoundRobin. The performance metrics are time, cost, and energy.
3) Optimized the decision support system (DSS) in the production line by implementing predictive maintenance.
4) Presented a detailed case study of manufacturing equipment-predictive maintenance using GA and two-class logistic regression.
5) Proposed promising future directions for this research article.

B. Article Organization

The remainder of this article is organized as follows. Section II presents the literature review of the existing techniques. Section III describes the proposed resource management technique. The experimental setup and case study are presented in Section IV. Section V presents the results of the evaluation. Section VI concludes the article and further work.

II. Related Works

The institute of asset management (IAM) built up a theoretical model for physical, virtual, and human asset management. IAM assembled the six key subjects for assets management, to be specific: 1) strategy and planning; 2) asset management decision making; 3) lifecycle delivery; 4) asset information; 5) organization and people; and 6) risk and reviews [8].

The authors proposed a novel classification for multiunit systems asset management into the fleet and portfolio according to a variety of assets and intervention options [9]. Fleet was referred to a system of homogeneous assets while portfolio was a system of the heterogeneous asset. The dependencies in multiunit systems are categorized into performance, stochastic, and resource. Then, the authors concluded that safety and reliability for multiunit systems were complex models that involved various criteria from various dimensions.

The service placement policy, namely, MinRE introduced with the aims to supply high QoS for IoT devices and to lower energy consumption in fog computing [10]. The authors classified the services into critical and normal. The goal of critical service was to reduce the responding time while normal service to lower energy consumption. MinRE organized the services in ascending order based on the deadline and priority was given based on the classified services. The policy was evaluated through a simulation experiment and the results evidenced that MinRE outperformed cloud-only, edge-ward, and resource-aware.

Denial-of-Service (DoS) attack prevention and energy conservation was an essential concern in IoT [11]. The received signal strength (RSS) was introduced to prevent DoS and conserved energy. RSS measured the receiving signal power to determine if the attacker on the same network using a teaching–learning-based optimization (TLBO) algorithm. Simulation results suggested that RSS was able to locate the attacker within 12 cm and the false alarm probability was 0.7%.

The load-balancing mechanism depends on the Jena architecture and contract-net protocol (CNP) to manage the smart manufacturing equipment at the floor level posited by Wan *et al.* [12]. First, the resource ontology model was introduced to collect and visualize the knowledge for sharing, reusing, and reasoning. Jena reasoning inputted the base model from ontology to extract the hidden information and decided the operating mode. CNP received the input from Jena reasoning to distribute the resources through three mechanisms: 1) open tender; 2) bidding; and 3) winning mode.

Containerization for resource allocation in the fog computing instead of virtual machine (VM) application is proposed by Yin *et al.* [13]. This was due to the container was more lightweight and had higher efficiency compared to VM. The concept was supported with simulation experiments, where the container outperformed the VM. The authors proposed a novel task scheduling algorithm based on threshold evaluation to amplify the jobs in fog nodes and to diminish the job delays. However, this research did not take consideration of the fitness of cloud resources and neglected the cloud computation time.

A lightweight architecture system with cloud and edge, namely, SERENA, to implement the predictive analytics platform [14]. SERENA collected the sensors data in the edge gateway and processed the information in the hybrid cloud. SERENA managed the deployment in Docker service and utilized the load balancing from Docker to distribute the tasks. SERENA enabled the predictive analytic service to examine the condition of the manufacturing equipment using three different machine learning algorithms: 1) decision tree; 2) gradient boosted tree; and 3) random forest.

A. Critical Analysis

Table I shows the comparison of the proposed resource management technique with the existing works. All of the current works only considered physical assets without considering

TABLE I
COMPARISON OF THE PROPOSED RESOURCE MANAGEMENT TECHNIQUE WITH EXISTING WORK

Work	**Asset**			**Equipment prediction maintenance**	**Fog/ cloud/ Local**	**Performance metrics**			**Method**
	Physical	Virtual	Human			Time	Energy	Cost	
[8] (Backman et al. 2016)	✓	✓	✓		Local				N/A
[10] (Hassan et al. 2015)	✓				Fog	✓	✓		Policy: MinRE
[11] Ghahramani et al. 2020)	✓		✓		N/A				N/A
[12] (Wan et al. 2018)	✓				Cloud		✓		Load balancing based on Jena reasoning and CNP
[13] (Yin et al. 2018)	✓				Fog	✓			Containerisation
[14] (Panicucci et al. 2020)	✓			✓	Fog				Load balancing
This work (proposed)	✓	✓	✓	✓	Fog	✓	✓	✓	GA-based

virtual and human assets except [8]. Physical, virtual, and human assets are equally crucial for the growth of business in Industry 4.0. A predictive maintenance model enables the team to fix the problem before equipment failure. However, only Panicucci *et al.* [14] highlighted equipment predictive maintenance and resource scheduling in their work. None of the existing literature considered all the three performances metrics: 1) time; 2) energy; and 3) cost, in their proposed resource management technique for physical, virtual, and human assets. Owing to the reasons mentioned above, the current literature becomes inefficient when solving the real-life Industry 4.0 manufacturing problem, where Industry 4.0 links automation, equipment, labor, and software altogether and requires fog computing for low latency.

Therefore, it is necessary to build up a resource management technique that comprises of physical, virtual, and human assets for Industry 4.0. Time, cost, and energy of the resource management technique should be taken into consideration. The industrial big data from IIoT sensors shall be fully utilized in the predictive maintenance application. This article addresses the challenges of the existing resource management technique.

III. PROPOSED TECHNIQUE

This section presents a detailed description of the proposed system architecture for the asset management and resource scheduling technique to handle the incoming tasks from IIoT sensors. Especially in this article, the question is being responded: how to manage the assets in Industry 4.0 effectively using GA and astutely utilize the industrial big data to minimize manufacturing equipment failure through supervised machine learning?

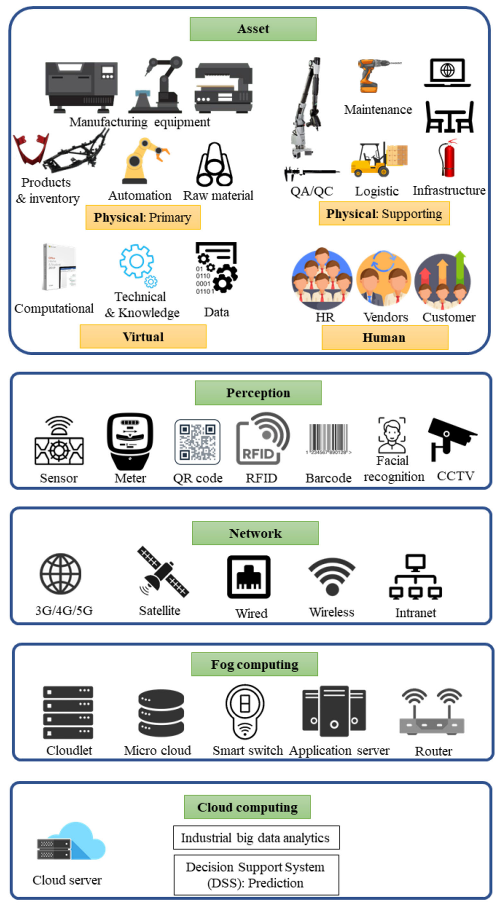

Fig. 1. System architecture.

A. System Architecture

Fig. 1 presents the proposed system architecture of assets management. The architecture consists of five layers: 1) asset; 2) perception; 3) network; 4) fog computing; and 5) cloud computing according to their functionalities.

The *asset layer* contains all the resources with economic values owned by the business with the expectation to produce value. The assets are identified as primary physical, supporting physical, virtual, and human. Primary physical assets are the central elements that are required for manufacturing and the manufactured products. The central elements are different manufacturing equipment and automation equipment that vary according to the nature of the industry. Supporting physical assets are the elements that enable and keep the primary manufacturing process going. Virtual assets enable digitalization in the business and manufacturing process through the integration of IT software. Humans, such as employees, vendors, customers, and end customers are the parties that directly involved in the life cycle of manufactured products. Employees are necessary to conduct manual operation, maintenance, and problem solving that cannot be replaced by automation equipment.

The *perception layer* consists of industrial smart sensors to gather the environmental and product information. Industrial smart sensors and meters installed in the equipment are able to detect and send physical parameters for prediction maintenance. Vision sensors are able to read the QR code and barcode, which contain vulnerable information, such as asset type, location, date of purchase, etc., about the assets. Facial recognition eases human resource management by reducing time fraud and employees' access control.

The *network layer* is in charge of transmitting the real-time data from sensors to network devices, fog computing, and computing layers. Business owners with multiple manufacturing plants in the whole world can be linked together to a global business through satellite communications. Wired, wireless, and Intranet connections allow the communication of assets within the business.

The *fog computing layer* creates communication between edge devices and the cloud data center [15]. Fog computing is a distributed decentralized system and allows the data to send to the server to process locally. Fog computing enables real-time asset analytic applications due to the nature of low latency and bandwidth connections compared to the cloud. Cloudlet and microcloud are small-scale data centers situated at the edge of the network. The smart switch allows digital facilities management to create sustainability and greener environment. Meanwhile, the application server allows the server to run industry-specified software individually. The router at fog computing levels enables several IIoT applications, such as data acquisition, smart metering, and distribution automation.

The *cloud computing layer* allows resources management, industrial big data, and IIoT task processing [15]. Preprocessing, training, testing, prediction, and model deployment of industrial big data are performed at this level due to the cloud offer flexibility: pay as you go. Cloud allows resource management and scheduling according to the policies of the business.

B. Scheduling: Genetic-Algorithm-Based Resource Scheduling Technique

Algorithm 1 shows the pseudocode for GA-based resource scheduling technique. GA mimics Darwin's theory of evolution, where the fittest survive in nature and GA works based on state-space search. In nature, the fitter organism has a higher survival rate and able to possess its genes to the next generation through reproduction. This allows the new and fitter generation better to adapt themselves in nature. GA starts with a set of variables, where the number of chromosomes denotes as population. Their fitness value assesses new solutions (offspring). GA utilizes three operators: 1) *selection*; 2) *crossover*; and 3) *mutation* to improve the solutions in each generation.

The *selection* includes picking the parents' feature vectors from the same generation as indicated by the fitness value. Meanwhile, the main objective of *crossover* is to ensure the parents' genes are exchanged and the offspring inherits the combined genes of parents. Zero cross rate implies no crossover has taken place and the offspring is an exact copy of parents. Crossover alone does not introduce a new feature vector to the offspring and possible to lead to similar solutions in the new generations. *Mutations* are introduced to cause random changes in the locus (positions in the chromosome). The offsprings are then being placed in the new generations until the end condition is fulfilled. GA can solve the resource scheduling problem due to the following characteristics.

1) GA is favorable in band selection, smaller size classification, training, and testing accuracies in contrast to ABC and PSO.
2) GA initializes the search from population points instead of a single point.
3) GA is a direct method for global search and thus avoids trapping in local optima.

Table II presents the detailed terminology used in this research article.

TABLE II
GA TERMINOLOGY

GA Terminology	Description
Population size	The number of job requests from IIot sensors and devices. Represented by the number of chromosomes in one iteration/ generation.
Number of iterations	The number of generations
Cross rate	The probability of accepting a new feature vector of a job
Mutation rate	The random probability where the elements inside the feature vectors are flipped or changed. Changes caused by errors while copying parents' feature vectors.
Gene	Contained inside the feature vector

Algorithm 1 GA-Based Resource Scheduling Technique

```
Begin
Input: utilisation metrics as feature vectors
Output: scheduling decision
generate n population
t=0
while (not terminating condition) do
begin
compute fitness function
t=t+1
select two parents' feaure vectors
apply crossover to feature vectors to generate offspring
apply mutation to offspring
replace previous population
end
return best offspring from population
end while
End
```

IV. PERFORMANCE EVALUATION

To show the practicality of the proposed strategy, this section executes and deploys a case study on the present reality physical asset: manufacturing equipment predictive maintenance on FogWorkflowSim[1] and Microsoft Azure Machine Learning Studio.

[1]FogWorkflowSim -https://github.com/ISEC-AHU/FogWorkflowSim.

A. Case Study: Manufacturing Equipment Predictive Maintenance

In the manufacturing industry, manufacturing machines, such as die casting machine, laser cut machine, plasma cutting machine, etc., are the essential equipment to produce goods for customers. However, unforeseen machine failure and components failure can lead to production line stoppage. The domino effects of unplanned production stoppage include delay in delivery, industrial consequences links to processes, and financial losses.

The arrival of Industry 4.0 and industrial big data has created a contemporary opportunity for manufacturing equipment predictive maintenance. Predictive maintenance utilizes the IIoT sensors to collect, evaluate, and analyze the real-time condition of the manufacturing equipment. The benefits of predictive maintenance include maintaining high overall equipment efficiency (OEE), early warning of anomalies, and awareness of the health condition of the equipment.

This case study runs on a desktop computer with configurations as described below.

1) *Processor:* Intel Core i9-9960X CPU@ 3.10 GHz.
2) *RAM:* 64 GB.
3) *System Type:* Windows 10 64-b OS.

B. Data Sets

The data sets used in this research article are available as follows.

Original (Highly Imbalance): https://bit.ly/2VyZY5i.

Amended (Undersampling Technique): https://bit.ly/37uVGS9.

The data sets used for this case study are created by Fidan Boylu Uz. Due to corporate confidentiality and intellectual property, the data of attributes are represented by general word and numerical number. The data set contains 11 attributes that identify the condition of manufacturing equipment: 1) *datetime:* recorded date and time; 2) *machineID:* physical assets identification number; 3) *errorID:* the error code; 4) *volt:* electrical voltage in Volt; 5) *rotate:* rotational speed; 6) *pressure:* measured pressure; 7) *vibration:* measured vibration; 8) *comp:* components replaced; 9) *model:* type of equipment; 10) *age:* age of the equipment; and 11) *failure:* 0 or 1. However, the data set is highly imbalanced, where it contains 6663 samples (2.28%) of class "1" and 285 006 samples (97.72%) of class "0." The imbalanced data set will result in a high-accuracy model even without training.

The data set is then processed with the undersampling technique to overcome the imbalance issue. The undersampling technique randomly reduces the majority class to match the number with minority class. Consequently, the total amount of samples has reduced to 14 482, where 6663 samples are of class "1" and 7819 samples are of class "0."

C. Implementation of the Proposed Techniques in FogWorkflowSim

In the FogWorkflowSim, the simulation environment considers four end devices (voltage sensors, pressure sensors, vibrational sensors, and rotational sensors), five fog nodes (cloudlet, micro cloud, smart switch, application server, and smart router) and one cloud server for this case study. The MIPS values and the execution cost of each device are as suggested in [16].

TABLE III
FOG ENVIRONMENT SETTING IN FOGWORKFLOWSIM

Parameters	**End Device**	**Fog Nodes**	**Cloud Server**
Number of devices	4	5	1
Million instructions per **second (MIPS)**	1,000	1,300	1,600
Execution Cost (C$)	0	0.48	0.96

The selection of the parameters, especially cross rate and mutation rate in the GA algorithm are problem dependent. The parameters shown in Table V are tuned carefully according to previous studies [16]–[19] to optimist the performance matrices, such as time, energy, and cost. The performance metrics are obtained from (1)–(6) [20], [21]. The population size and number of iterations are fixed to keep the computational time low.

$$\text{Execution Time } t : t_i^{\text{total}} = t_i^{\text{tran}} + t_i^{\text{exe}} + t_i^{\text{rec}} + t_i^{\text{mig}}. \quad (1)$$

t_i^{tran} denotes the transmission time from end devices to fog server, t_i^{exe} represents the execution time at the fog server, t_i^{rec} is the transmission time from the fog server to end device, and t_i^{mig} refers to the task migration time to a different fog server because of the motion of the end device.

$$\text{Cost, } C : C_i^{\text{total}} = T_i^{\text{fog}} \times C^{\text{fog}} + T_i^{\text{cloud}} \times C^{\text{cloud}} \quad (2)$$

where T_i^{fog} is the workflow task at the fog server, T_i^{cloud} is the workflow task at the cloud server, C^{fog} denotes the unit price per second in the fog server, and C^{cloud} represents a unit price per second in cloud computing.

$$\text{Energy Usage } E : E_i^{\text{total}} = E_i^x + E_i^y + E_i^z \quad (3)$$

$$E_i^x = \frac{\text{Data transmission}}{\text{Bandwith}} \times P_{\text{transmission}} \quad (4)$$

$$E_i^y = \frac{\text{Task workload}}{\text{Task processing speed}} \times P_{\text{idle}} \quad (5)$$

$$E_i^z = \frac{\text{Task workload}}{\text{Task processing speed}} \times P_{\text{end}}. \quad (6)$$

E_i^x represents transmission energy from end devices to fog server, E_i^ydenotes idle energy utilisation of the end devices, and E_i^z is the load energy utilisation.

The purpose of the scientific workflow is to show the dependencies between tasks and manage the data flow. Montage workflow with 60 jobs is found to be able to optimize the performance matrices.

Table III illustrates the configuration settings in the FogWorkflowSim. Table IV shows the GA settings, while Table V describes the type of workflow and total job.

D. Two-Class Logistic Regression Equipment Predictive Maintenance in Microsoft Azure Machine Learning

The purpose of this case study is to predict the condition of manufacturing equipment using the two-class logistic

TABLE IV
GA SETTING IN FOGWORKFLOWSIM

Parameters	**Value**
Population size	50
Number of iterations	100
Cross rate (%)	85
Mutation rate (%)	1

TABLE V
WORKFLOW SETTING IN FOGWORKFLOWSIM

Parameters	**Input**
Workflow type	Montage
Total job	60

TABLE VI
PARAMETERS FOR TWO-CLASS LOGISTIC REGRESSION IN 70:30 DATA SPLITTING

Parameters	**Values**
Optimisation tolerance	0.000100009
L1 **Regularisation** weight	0.10009
L2 **Regularisation** weight	0.10009
Memory size **(MB)**	11
Quiet	True
Use threads	True
Allow unknown levels	True
Random number seed	12345

regression algorithm, where 0 implies healthy equipment; meanwhile, 1 denotes equipment failure. The data set is obtained from Section IV-B. The attributes that determined the failures are: errorID, volt, rotate, pressure, vibration, comp, and age. The data set is divided into training and testing sets in the ratio of 70:30. The training data set is used for building up the prediction algorithm while the testing data set enabled the model assessment on the real-time data. Table VI describes the best parameters for two-class logistic regression after tuning. For deployment, the maintenance team can predict the equipment condition based on the real-time data input through RESTful API or Microsoft Excel. The complete workflow for this case study is available at https://bit.ly/2I5uP6u.

V. EXPERIMENTAL RESULTS

A. Performance of GA in FogWorkflowSim

Fig. 2 compares the: 1) execution time; 2) operating cost; and 3) energy usage of GA with MinMin, MaxMin, FCFS, and RoundRobin scheduling algorithm.

1) *Execution Time:* The execution time of GA is 28.1%, 9.1%, 0.5%, and 9% faster compared to MinMin, MaxMin, FCFS, and RoundRobin, respectively. The execution time of GA is much lower due to the flexibility of parameter tuning. Crossover is the most important operation and enables the good characteristics of the individual parents to recombine. The algorithm can discover the solution more efficiently throughout the acceleration in each generation and has the lowest execution time among all. Furthermore, the execution time of GA is lower than the heuristic algorithm, e.g., RoundRobin despite the complexity of GA is due to job scheduler in CloudSim. When the user sends a high number of job requests to the cloudlet simultaneously, CloudSim is not able to execute them and job scheduler places the requests in a queue system. The job scheduler positions and executes the job request from the lowest execution time to the highest and sends back to the user. This leads to lower execution times of the GA method.

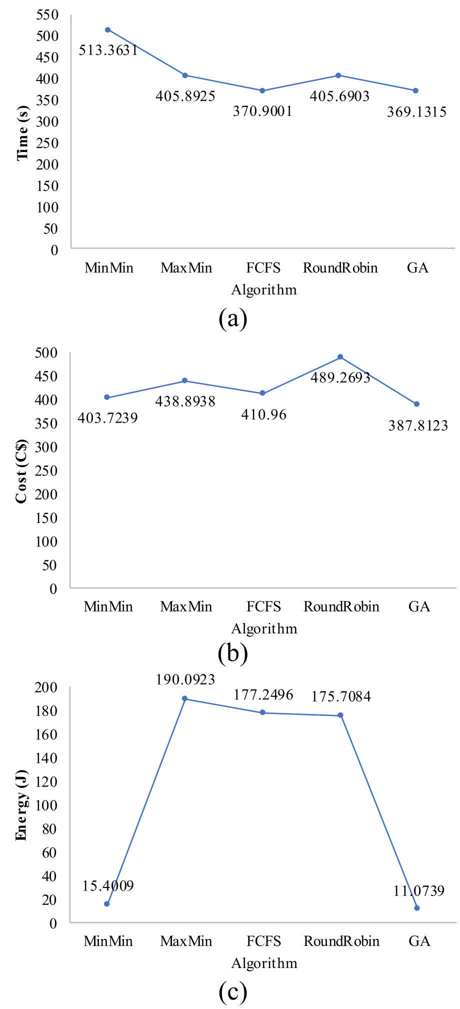

Fig. 2. Evaluation results for MinMin, MaxMin, FCFS, RoundRobin, and GA: (a) time, (b) cost, and (c) energy.

2) *Cost:* The cost of GA is 28.1%, 94.2%, 93.8%, and 93.7% lower compared to MinMin, MaxMin, FCFS, and RoundRobin, respectively. GA optimizes the distribution of tasks and has a better fitness value. Thus, this further reduces the executing cost.

3) *Energy Usage:* The energy usage of GA is 3.9%, 11.6%, 5.6%, and 20.7% lesser compared to MinMin, MaxMin, FCFS, and RoundRobin, respectively. Montage workflow allows label-based clustering, where the same tasks in the workflow can cluster together. Instead of executing individual workload, clustered workload minimizes the network traffic and further reduces energy usage. Furthermore, the GA strategy tends to club computationally expensive tasks in resource-intensive cloud nodes and simpler tasks in edge devices. This leads to a reduction in energy usage.

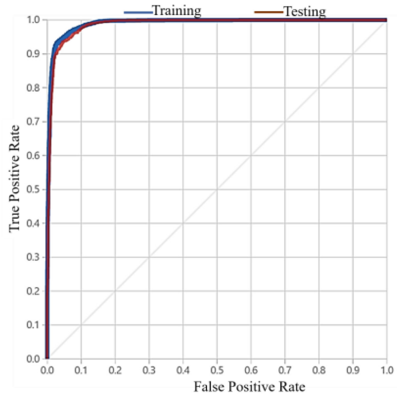

Fig. 3. Training and testing of the two-class logistic regression ROC curve.

TABLE VII
CONFUSION MATRIX AND MEASURE OF TRAINING AND TESTING

Training	TP 4412	FN 274	Accuracy 0.951	Precision 0.952	Threshold 0.5
	FP 223	TN 5228	Recall 0.942	F1 Score 0.947	AUC 0.990
Testing	TP 1844	FN 133	Accuracy 0.945	Precision 0.946	Threshold 0.5
	FP 105	TN 2263	Recall 0.933	F1 Score 0.939	AUC 0.987

B. Training and Testing Accuracy of Predictive Maintenance

The equipment predictive maintenance employs two-class logistic regression to predict the health of the manufacturing equipment. Two-class logistic regression allows good accuracy with fast training time [22]. Fig. 3 depicts the receiver operating characteristic (ROC) curve of training and testing data sets. Setting the threshold as a constant value of 0.5 in both training and testing data sets, the area under curve (AUC) of ROC is 0.990 and 0.987, respectively. AUC closer to 1 represents a better measure of separability, whist AUC = 1 is where classifier impeccably recognizes all positive and negative classes accurately. The training accuracy and testing accuracies of the model are 95.1% and 94.5%, respectively. Table VII shows the confusion matrix along with their measures of training and testing data sets.

VI. CONCLUSION

This article proposed the GA as the technique for resource management in asset management application for Industry 4.0. The proposed system architecture contains five layers, including: 1) assets; 2) perception; 3) network; 4) fog computing; and 5) cloud computing. GA was evaluated along with MinMin, MaxMin, FCFS, and RoundRobin in FogWorkflowsim to show the effectiveness of the proposed technique. The performance metrics for the evaluation were execution time, cost, and energy. Extensive simulation experiment evidenced that GA outperformed MinMin, MaxMin, FCFS, and RoundRobin in terms of having the lowest execution time, cost, and energy. The execution time was 0.48% faster, the cost was 5.43% lower, and energy usage was 28.10% lower in comparison to second-best results. Finally, a model for equipment predictive maintenance had been deployed using a supervised machine learning algorithm, two-class logistic regression. The model was able to predict if the manufacturing equipment failing and produced an early warning alert for the production line. The training accuracy and testing accuracy for the model were 95.1% and 94.5% each.

A. Future Work

In spite of the fact that the proposed resource management technique demonstrated efficiency and able to distribute the tasks effectively, it very well may be additionally enhanced in a broader scope followed by the accompanying viewpoints.

1) *Reliability and Security Communication:* There is a need to ensure up-to-date industrial security communication among the devices to prevent cyberattacks.
2) *Performance Metrics:* The performance metrics of the simulation can further include network latency, network bandwidth, and jitter.
3) *Extending to Varies Domains:* The current article focuses on the manufacturing industry. This can be extended to various domains of Industry 4.0, such as construction, oil and gas, chemical due to the beneficial of asset usage, quality control, supply chain management, product monitoring, and work environment wellbeing.

REFERENCES

[1] S. S. Gill *et al.*, "Transformative effects of IoT, blockchain and artificial intelligence on cloud computing: Evolution, vision, trends and open challenges," *Internet Things*, vol. 8, Dec. 2019, Art. no. 100118.
[2] M. S. de Brito, S. Hoque, R. Steinke, A. Willner, and T. Magedanz, "Application of the fog computing paradigm to smart factories and cyber-physical systems," *Trans. Emerg. Telecommun. Technol.*, vol. 29, no. 4, pp. 1–14, 2018.
[3] D. Mourtzis, E. Vlachou, and N. Milas, "Industrial big data as a result of IoT adoption in manufacturing," *Procedia CIRP*, vol. 55, pp. 290–295, Feb. 2016.
[4] S. Tuli *et al.*, "HealthFog: An ensemble deep learning based smart healthcare system for automatic diagnosis of heart diseases in integrated IoT and fog computing environments," *Future Gener. Comput. Syst.*, vol. 104, pp. 187–200, Mar. 2020.
[5] S. Poslad, *Ubiquitous Computing*. Chichester, U.K.: Wiley, 2009.
[6] S. Singh and I. Chana, "A survey on resource scheduling in cloud computing: Issues and challenges," *J. Grid Comput.*, vol. 14, no. 2, pp. 217–264, Jun. 2016.
[7] J. Alcaraz and C. Maroto, "A robust genetic algorithm for resource allocation in project scheduling," *Ann. Oper. Res.*, vol. 102, nos. 1–4, pp. 83–109, 2001.
[8] J. Backman, J. Vare, K. Framling, M. Madhikermi, and O. Nykänen, "IoT-based interoperability framework for asset and fleet management," in *Proc. IEEE Int. Conf. Emerg. Technol. Factory. Autom. (ETFA)*, vol. 2016, Nov. 2016, pp. 1–4.
[9] S. Petchrompo and A. K. Parlikad, "A review of asset management literature on multi-asset systems," *Rel. Eng. Syst. Safety*, vol. 181, pp. 181–201, Jan. 2019.
[10] H. O. Hassan, S. Azizi, and M. Shojafar, "Priority, network and energy-aware placement of IoT-based application services in fog-cloud environments," *IET Commun.*, vol. 14, no. 13, pp. 2117–2129, Aug. 2020.
[11] M. Ghahramani, R. Javidan, M. Shojafar, R. Taheri, M. Alazab, and R. Tafazolli, "RSS: An energy-efficient approach for securing IoT service protocols against the DoS attack," *IEEE Internet Things J.*, early access, Sep. 10, 2020, doi: 10.1109/JIOT.2020.3023102.
[12] J. Wan *et al.*, "Toward dynamic resources management for IoT-based manufacturing," *IEEE Commun. Mag.*, vol. 56, no. 2, pp. 52–59, Feb. 2018.

[13] L. Yin, J. Luo, and H. Luo, "Tasks scheduling and resource allocation in fog computing based on containers for smart manufacturing," *IEEE Trans. Ind. Informat.*, vol. 14, no. 10, pp. 4712–4721, Oct. 2018.

[14] S. Panicucci *et al.*, "A cloud-to-edge approach to support predictive analytics in robotics industry," *Electronics*, vol. 9, no. 3, pp. 1–22, 2020.

[15] S. S. Gill, P. Garraghan, and R. Buyya, "ROUTER: Fog enabled cloud based intelligent resource management approach for smart home IoT devices," *J. Syst. Softw.*, vol. 154, pp. 125–138, Aug. 2019.

[16] X. Liu *et al.*, "FogWorkflowSim: An automated simulation toolkit for workflow performance evaluation in fog computing," in *Proc. 34th IEEE/ACM Int. Conf. Autom. Softw. Eng. (ASE)*, 2019, pp. 1114–1117.

[17] M. Obitko. (1998). *Introduction to Genetic Algorithms*. Accessed: Aug. 1, 2020. [Online]. Available: https://courses.cs.washington.edu/courses/cse473/06sp/GeneticAlgDemo/main.html

[18] M. Birattari, *Tuning Metaheuristics*, vol. 197, 197th ed. Berlin, Germany: Springer, 2009.

[19] H. M. Pandey, A. Chaudhary, and D. Mehrotra, "A comparative review of approaches to prevent premature convergence in GA," *Appl. Soft Comput. J.*, vol. 24, pp. 1047–1077, Nov. 2014.

[20] L. Fan, X. Liu, X. Li, D. Yuan, and J. Xu, "Graph4Edge: A graph-based computation offloading strategy for mobile-edge workflow applications," in *Proc. IEEE Int. Conf. Pervasive Comput. Commun. Workshops (PerCom Workshops)*, 2020, pp. 3–6.

[21] J. Xu *et al.*, *Mobility-Aware Workflow Offloading and Scheduling Strategy for Mobile Edge Computing* (Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)), vol. 11945. Cham, Switzerland: Springer, 2020, pp. 184–199.

[22] C. J. Gronlund, J. Martens, and F. Lazzeri. (Jul. 5, 2020). *How to Select a Machine Learning Algorithm*. Accessed: Aug. 10, 2020. [Online]. Available: https://docs.microsoft.com/en-us/azure/machine-learning/how-to-select-algorithms

Yyi Kai Teoh received the B.Eng. (Hons.) in mechanical engineering from the University of Nottingham, Nottingham, U.K. He is currently pursuing the M.Sc. degree with the Queen Mary University of London, London, U.K.

He is currently completing his industrial placement at an IoT consulting company in London, working as IoT and Hardware Systems Analyst. His research interests include IoT, machine learning, distributed systems, and cloud computing.

Sukhpal Singh Gill received the Master's of Engineering degree (Gold Medalist) in software engineering, and the Doctoral degree in autonomic cloud computing from the Thapar University, Patiala, India.

He is a Lecturer (Assistant Professor) in Cloud Computing with the School of EECS, Queen Mary University of London, London, U.K. Prior to this, he has held positions as a Research Associate with the School of Computing and Communications, Lancaster University, Lancashire, U.K. and also as a Postdoctoral Research Fellow with the CLOUDS Laboratory, University of Melbourne, Melbourne, VIC, Australia. His research interests include cloud computing, fog computing, Internet of Things, and energy efficiency.

Ajith Kumar Parlikad received the Ph.D. degree from Cambridge University, Cambridge, U.K., in August 2006.

He is a Reader in Asset Management with the Engineering Department, Cambridge University. He is based at the Institute for Manufacturing, where he is the Head of the Asset Management Research Group, Cambridge University, where he is a Fellow and a Tutor with Hughes Hall. His current research focusses on the development and exploitation of digital twins of complex asset systems bringing together data from disparate sources to improve asset management.